

Towards using Involuntary Body Gestures for Measuring the User Engagement in VR Gaming

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ABSTRACT

Understanding the degree of user engagement in a VR game is vital to provide a better gaming experience. While prior work has suggested self-reports, and biological signal-based methods, measuring game engagement remains a challenge due to its complex nature. In this work, we provide a preliminary exploration of using involuntary body gestures to measure user engagement in VR gaming. Based on data collected from 27 participants performing multiple VR games, we demonstrate a relationship between foot gesture-based models for measuring arousal and physiological responses while engaging in VR games. Our findings show the possibility of using involuntary body gestures to measure engagement.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

VR, VR Games, Gaming, Play, Body Gestures, IMU, Foot Gestures, Emotions, Component Process Model, Stress

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1 INTRODUCTION & RELATED WORK

Billions of people participate in digital gaming from casual games to real-time strategy or shooters to multi-player games. With the advancement of VR, gaming has further grown as an engaging forms of entertainment and has become a platform in which to be immersed. Unlike conventional gaming, VR games can keep users deeply engaged with the gaming activity with fewer distractions



Figure 1: The figure shows some images of the study setup. Participants wore an IMU near right ankle.

from the real world. While immersion has been defined as one's degree of involvement [2], engagement is considered the first step towards immersion [1, 2]. Hence, measuring the degree of engagement is important to provide a better VR gaming experience.

According to the literature, engagement can be defined as the attention that someone gives to a particular activity, object, or person that can be measured as the increase of arousal or attention [11]. Hence, prior work has suggested methods to measure engagement, including self-reported methods [9], physiology-based methods such as measuring skin conductance [10], and behaviour-based methods such as mouse movements [3, 11] or intensity of keyboard pressure [8]. Moreover, prior work suggests that measuring game engagement using multiple methods will work better due to the complex nature of engagement [11]. Hence, in this work, we propose using involuntary body gestures to measure user engagement in VR gaming as a modality that could complement the existing methods of measuring engagement. According to our knowledge, this work can be considered the first to explore measuring engagement in VR gaming using involuntary body gestures.

As a preliminary exploration, we demonstrate the possibility of using involuntary foot gestures in measuring engagement in VR gaming. First, in a laboratory study, we collected self-reports, EDA, and foot motion data, while participants performed 27 VR games in 3 sessions. Then using a publicly available data set [4, 5], we developed a model that can measure arousal using involuntary foot gestures and utilized those to infer cognitive stress based on the involuntary foot posture and motion characteristics. Later, we analysed the correlation between the model-derived stress levels and the collected self-reports and EDA data. Results show a significant correlation between derived stress levels and self-reported data that can be used to measure engagement. In summary, our results

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demonstrate the possibility of using body gestures as a method of measuring engagement to complement traditional methods of measuring engagement.

2 METHODOLOGY

2.1 Model Development

We aim to investigate the potential of using involuntary body gestures to measure user engagement while playing VR games. According to prior work, engagement is related to elevated attention and arousal [7, 11] which indicates increased cognitive stress. Hence, in this preliminary study, we aim to investigate the possibility of using body gestures to detect cognitive stress while performing VR games. Anecdotal evidence [12, 13] and research in literature [4, 6] shows the foot as an interesting location to sense body gestures. Moreover, prior work estimates cognitive stress in performing sedentary computer tasks based on involuntary foot gestures [4]. Hence, in this preliminary study, we adopted a model based on prior work [5] and used their publicly available data set to derive a model that can detect cognitive stress based on involuntary foot gestures. Our study was also performed in a seated position, and we utilised similar sensors (SAM21 board with 9DOF Inertial Measurement Unit (IMU) - MPU 9250) with similar settings. Hence, the data we collected while playing VR games can be directly fed to the stress estimation model. We used four features related to foot posture aZ_{mean} (Z axis acceleration), aX_{mean} (X axis acceleration), aY_{mean} (Y axis acceleration) and motion F_M (median frequency) and trained linear discriminant analysis model (LDA), which resulted in the decision rule: $LD1 = 0.9387970 * aZ_{mean} + 0.1518052 * aY_{mean} - 0.1146619 * aX_{mean} - 0.2839574 * F_M$. The decision rule estimated whether the user shows stressed or relaxed gestures every 20-second window. Based on the total number of windows showing stress posture, the stress level was calculated as $S_{Level} = (nWindows_{Stressed} / nWindows_{Total}) * 100$.

2.2 Data Collection & Analysis

In this preliminary study, we used 27 interactive VR games pre-labeled by a dominant emotion [17]. We used an HTC Vive Pro headset to deliver the VR experience to 27 participants (12 females, mean age = 24.4 (SD = 5.2) years) who played 27 games in 3 sessions in a seated position. Also, we attached the IMU sensors to the participant's right foot, as shown in the Figure 1. Participants were also asked to wear an E4 wristband on their non-dominant hand to collect skin conductance data. In addition, we randomised the game order across participants and sessions by the previously labeled descriptors. Each game lasted for three minutes, and after each game, the participants were given the Geneva Emotion Wheel (GEW) [16] and the CoreGRID questionnaires [15]. In GEW participants self reported about 20 emotions in a 5 point Likert scale (1 - *Not at All*, 5 - *Strongly Agree*). In the CoreGRID questionnaire, participants were asked to self-report their experiences described by the five main components of the Component Process Model (appraisal, motivation, expression, physiology, and feeling) [14] on a 5-point Likert scale (1 - *Not at All*, 5 - *Strongly Agree*). In our analysis, first, we derived the stress levels for each game (See Figure 2). Then we analyzed for any correlations between the derived stress level and components in GEW and CoreGRID questionnaires. Also,

we checked for correlations between derived stress levels and skin conductance levels (SCL).

3 FINDINGS & DISCUSSION

According to derived stress levels, 'Cosmic Flow', a relaxation VR experience, showed the lowest stress level indicating lower arousal and higher relief which was matched with self-reports of participants as well. A highly engaging VR game 'Ocean Rift' with a shark attack scenario, and a deep-sea experience, showed a higher stress level than relaxing VR experiences such as 'Cosmic Flow' or 'Meditation VR.' A one-way Anova test and a Tukey's HSD range test confirmed that the stress level occurred by 'Ocean Rift' is significant compared to 'Cosmic Flow' ($F_{26,677}=1.933$, $p < .01$). This shows that the stress detection model that uses involuntary foot gestures has successfully derived the expected stress levels, demonstrating the potential of measuring engagement using involuntary foot gesture models. Moreover, physiological signals have significantly correlated with the derived stress levels. In our analysis, we calculated the mean SCL difference between each game's first minute and last minute. The results showed a significant positive correlation ($p < .05$, $r = 0.45$). In CoreGRID questionnaire majority of the questions related to change of feelings, physiology responses and expressions showed a significant correlation. For instance, CoreGRID responses for experiencing prolonged heightened emotional state was statistically significantly correlated ($p < .05$) with the derived stress level with a correlation coefficient $r = 0.08$. Also the questions about the calmness had a statistically significant negative correlation ($p < .05$, $r = -0.12$). The physiology related questions such as "Did your stomach clench up" ($p < .05$, $r = -0.075$), "Did you experience muscle tension" ($p < .05$, $r = -0.098$), "Did you feel your breathing slowing down" ($p < .05$, $r = -0.079$) demonstrated a statistically significant correlation. Also, most of the self-rating for positive emotions on the GEW questionnaire was negatively correlated with the model-derived stress levels. For example, relief showed a statistically significant negative correlation ($p < .05$, $r = -0.23$). In summary, all of the above evidence shows that involuntary foot gestures can detect cognitive arousal, indicating high attention. In other words, this shows the possibility of using involuntary body gestures in measuring user engagement while performing VR games.

4 LIMITATIONS, FUTUREWORK & CONCLUSION

The current model was initially used to detect cognitive stress in computer-based sedentary tasks. We anticipate better results with a customised model for VR gaming. The current model will only work in the sitting posture since the study was conducted in a seated position. In the future, we aim to develop a model that helps measure engagement using full-body movements. For that, we could utilise motion sensors attached to shoes, motion sensors of the head-mounted display, and hand controllers. A vision-based system is another alternative, as new VR devices come with video capture. In conclusion, our work demonstrate the preliminary exploration of using involuntary body gestures in measuring user engagement.

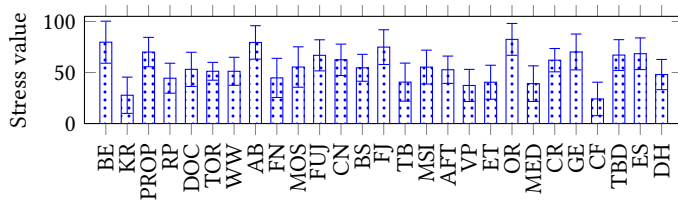


Figure 2: Stress levels for each game

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